

Creativity and Innovation

Creativity: It is described as bringing a unique approach and way of thinking to the way you do the things.

Consider the two examples given in Fig.1. The pill pack is having a medicines in a container or a pack with a chart printed on them. This chart shows the time and frequency of using this medicines. This will help the senior citizens who often find it difficult to take medicines at the appropriate time. The other product shows a toothbrush that will automatically brush the teeth when someone keeps it in the mouth and presses the operating button. This is more helpful to the kids who find it difficult to brush their teeth. These two are the examples how to use creativity to bring effective solutions to some day-to-day problems.

	
Pill pack – Lists the medicines with the time of taking them – suitable for senior citizens	Automatically brushes teeth by moving either way – suitable for kids

Fig.1: Creative thinking ideas

Innovation: It is the creation of new products, techniques, processes and services by generating and implementing new ideas to make them useful. Solutions should be more practical and effective.

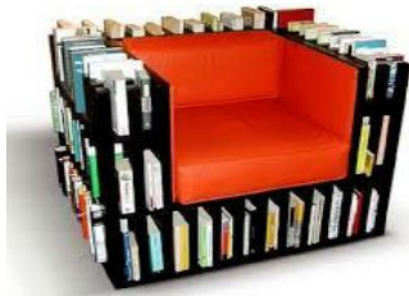
	
Laptop table – with shelves	Library seating chair – for study & reference

Fig.2: Innovative ideas

Creativity and Innovation

Consider the ideas illustrated in Fig.2. The laptop table is fitted with a simple and unique design of shelves to keep the books and other material. This design saves space and is more attractive. A chair is made with a number of shelves for keeping reference books for comfortable reading. This saves time and space for the reader without any need to fetch the books for other place for reading. The other innovations that are more prominent are – foldable bike and bracelet computers. Creativity is the thought process while Innovation is the execution of the creative ideas.

Art of Innovation

- The process of using creativity and new ideas to create practical and effective solutions for problems.
- It's a human-centered approach that involves understanding users' needs and generating ideas for new products, services, or processes

Differences between Creativity and Innovation

S No	Creativity	Innovation
1	Based on imagination	Productive work that has nothing to do with imagination
2	Thinking something new	Introducing something new
3	New ideas and put them into reality	Creative ideas put into practice
4	Innovative process	Productive process
5	Cannot be measured	Measurable
6	Does not require money	Requires money
7	No risk involved	Risk involved

Types of innovation: There are four types of innovations-

- **Product / Service Innovation**
 - Introduction of a goods or service that is new or significantly improved with respect to its characteristics or intended uses.
 - Includes significant improvements in technical specifications, components and materials, incorporated software, user friendliness or other functional characteristics.
 - Can utilize new knowledge or technologies, or can be based on new uses or combinations of existing knowledge or technologies.
- **Process Innovation**
 - Implementation of a new or significantly improved production or delivery method.
 - Significant changes in techniques, equipment and/or software.
 - Can be intended to decrease unit costs of production or delivery, to increase quality, or to produce or deliver new or significantly improved products.
- **Marketing Innovation**
 - Implementation of a new marketing method involving significant changes in product design or packaging, product placement, product promotion or pricing.

Creativity and Innovation

- Aimed at better addressing customer needs, opening up new markets, or newly positioning a firm's product on the market, with the objective of increasing the firm's sales.
- **Organizational Innovation**
 - Implementation of a new organizational method in the firm's business practices, workplace organization or external relations.
 - Can be intended to increase a firm's performance by reducing administrative costs or transaction costs, improving workplace satisfaction (and thus labour productivity), non-codified gaining access to non-tradable assets (such as external knowledge) or reducing costs of supplies.

Role of creativity and innovation in organizations

Creativity and innovation are essential for organizations to remain competitive, adapt to change, and drive growth. While they are related, they are not the same:

1. **Creativity** refers to the generation of new and original ideas. It involves thinking outside the box, problem-solving, and exploring new possibilities.
2. **Innovation** is the process of turning creative ideas into practical solutions, products, or services that add value.

The Importance of Creativity and Innovation in Organizations

1. **Competitive Advantage** – Organizations that foster innovation can differentiate themselves in the market.
2. **Adaptability** – Helps businesses respond to changes in technology, consumer demands, and economic shifts.
3. **Employee Engagement** – Encouraging creative thinking makes employees feel valued and motivated.
4. **Efficiency & Productivity** – Innovative processes and technologies improve workflow and reduce costs.
5. **Customer Satisfaction** – New and improved products enhance customer experiences.

Creative Thinking Tools:

1. Brainstorming

- This method is designed to encourage team members to express their thoughts in a free-flowing, spur-of-the-moment manner, building upon each other's ideas. One idea can spark another, creating a chain of insights.
- Healthy brainstorming focuses on quantity over quality. The goal here isn't just great ideas; it's loads of ideas. Outstanding concepts often surface from a sea of suggestions. Every thought, no matter how outlandish, is warmly welcomed.

Examples

- **Product Idea:** Participants generate as many ideas as possible for a new product without judging the feasibility or practicality of the ideas at this stage.



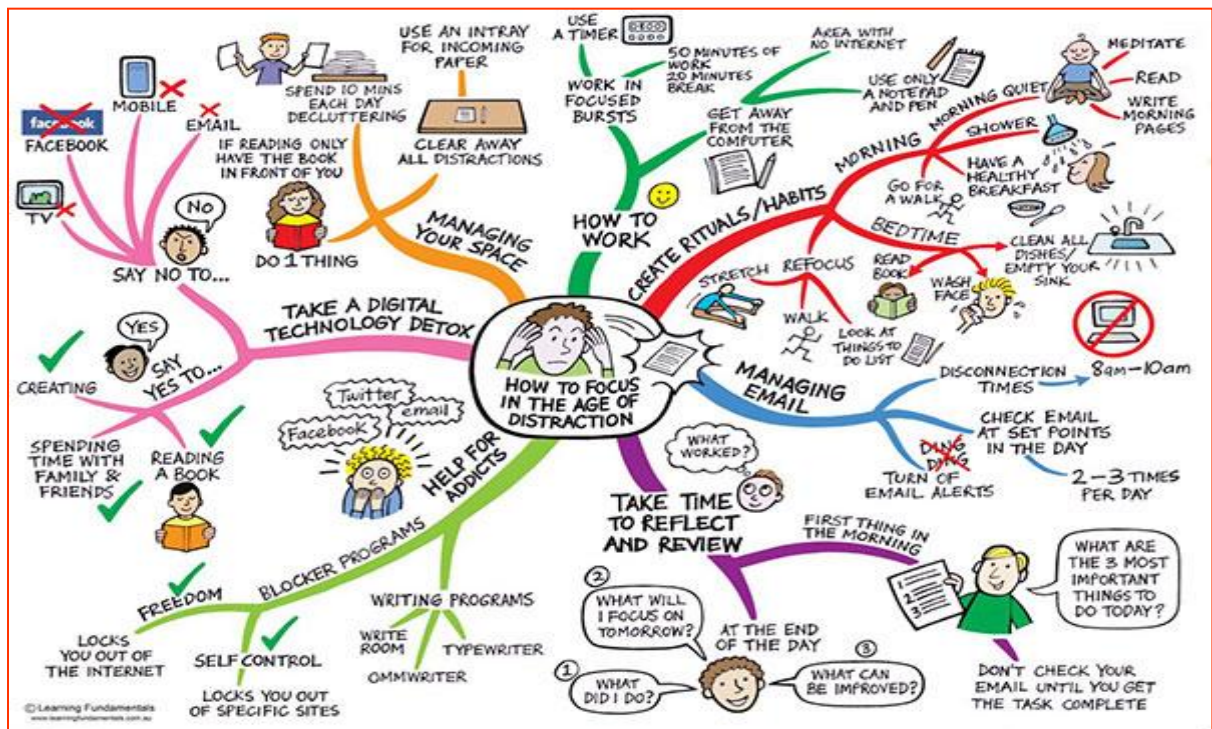
- **Fitness case:** Gather as many as possible ideas for the right combination of exercises or fitness activities for effective result.

2. Mind mapping:

- It is a graphical way to represent ideas and concepts, and their interrelationships, centering around a core subject.
- Imagine it as a tree with many branches. The central idea questions the trunk, and the branches are the primary sub-ideas, following branches being the subdivisions, and so on.
- Its visual approach helps to organize complex information, enabling efficient solution formations and decision making. It's a powerful tool for note-taking, memory enhancement, and creative idea generation.

Examples

- An entrepreneur may use a mind map to break down a business plan, visually representing areas like marketing, product development, and financial planning



3. Analogical Reasoning:

- It allows people to apply familiar experiences to new, unfamiliar situations, which stimulates new ideas and insights.

Examples

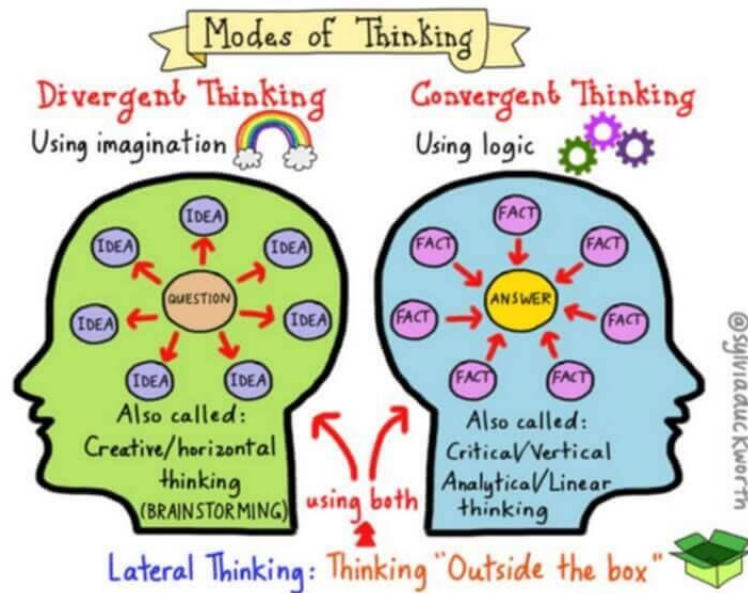
- A business facing budget allocation issues could use analogical thinking and look at how a household manages finances, utilizing that as a model for distributing the company's resources.

4. Lateral Thinking:-

- By breaking free from traditional thinking patterns, lateral thinking facilitates the generation of creative solutions. It encourages you to look at problems from different perspectives. It promotes **Out-of-Box thinking**.

Examples

- An ad company facing difficulties catching consumer attention might use lateral thinking to come up with a unique, unexpected advertising strategy, straying from the typical marketing conventions to better engage viewers.



5. Three Ifs...

The Three Ifs technique is about stimulating creativity through hypothetical 'If' scenarios.

- Imagine what could happen if certain variables in the current scenario were altered. Postulate three 'If' statements, and explore possible outcomes, thereby developing fresh perspectives.
- This technique fosters innovation and flexibility. It challenges people to break away from current limitations and imagine a different reality.

Examples

- A product designer could use 'If' scenarios like 'What if the product was half its size?', 'What if it doubled in functionality?' or 'What if it could be used in a totally different manner?' to create innovative designs.

6. Role Playing:

Role-playing is a method wherein individuals take on different personas to gain a fresh perspective on a situation or problem.

Creativity and Innovation

- Imagine wearing somebody else's shoes. By temporarily adopting another person's role, you can gain fresh insights and viewpoints.
- Role-playing can lead to empathetic understanding, improved communication, and innovative thinking. It helps examine situations through an additional lens.

Examples

- A team could employ role-playing to better understand customer needs and perspectives, with different team members taking on roles such as the customer, salesperson, and customer service representative

Creativity to Innovation:

1. **Idea Generation (Creativity)**
 - Encourages divergent thinking, brainstorming, and imagination.
 - Involves exploring new perspectives and questioning assumptions.
 - Techniques: Mind mapping, SCAMPER, lateral thinking, role-playing.
2. **Idea Selection & Development**
 - Evaluates ideas based on feasibility, viability, and desirability.
 - Prioritizes the most promising concepts for further exploration.
 - Uses feedback and collaboration to refine ideas.
3. **Prototyping & Experimentation**
 - Transforms ideas into tangible prototypes or MVPs (Minimum Viable Products).
 - Tests concepts with real users to validate their effectiveness.
 - Iterates rapidly to improve and optimize solutions.
4. **Implementation (Innovation)**
 - Deploys the refined idea into the market or organization.
 - Integrates business models, technology, and strategies to ensure success.
 - Measures impact and continuously adapts for improvement.

Value Based Innovation

It is a management approach that uses values to guide innovation. It considers values at the individual, organizational, societal, and global levels. The value of a product depends on its quality. Quality of a product depends on its price. The relation among them can be represented mathematically as -

$$\text{Value} = \frac{\text{Quality}}{\$}$$

For example, consider a smart phone – a Chinese product and a high quality Samsung phone. The Chinese product may offer more or less similar features than of that a Samsung phone. Its working and safety are unpredictable. The Chinese phone may be available at a cheaper cost than that of a Samsung phone. Its value is definitely lesser than that of a Samsung phone, which is highly rated, allows safety features and data protection/updating features.

Creativity and Innovation

The steps involved in the process are

- **Identify the values:** Understand the values that motivates the people and organizations
- **Integrate Stakeholders:** Involve stakeholders in the Innovation process
- **Costs:** Consider the cost of innovation during the development
- **Create Value:** Create products or services that provide value to customers

Examples

- E-scooters that are lighter, smaller, and attractive. They are cheaper than conventional scooters in mileage and maintenance
- 24-hour TV network – more coverage and more profits through advertisements and viewership
- Cheaper flight services attracted more public and increased profits through more occupancy
- Health sector: Assessment of which treatment is most valuable for a particular patient through testing using specialized analytical instruments - reducing unnecessary cost and increasing reliability on medical services

Benefits:

- Can help organizations create innovations that matter
- Can help organizations survive in critical periods
- Can help the meet the demands of customers
- Can help in designing products that cost effective and have higher value in its function